

E7522502_5

The equation $2x^4 + xy + y^3 - y + 1 = 0$ has no integer solutions

Size $h = 47$, length $l = 11$

Theorem 1. *The equation*

$$2x^4 + xy + y^3 - y + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), with $(d_1, d_2) = (-11, 22)$, and the fact that an odd prime p with $(-d_k/p) = -1$ divides $U_k^2 + d_k V_k^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm forms $a^2 + db^2$

Throughout this section, d is a squarefree integer with $d \neq 0, -1$; we apply the results with $d \in \{-11, 22\}$. For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right) \left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(-d/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + db^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv -d(bc)^2 \equiv -d \pmod{p}$, so $-d$ is a square modulo p . As $(-d/p) \neq 0$, it is a non-zero square, so $(-d/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 + db^2 \neq 0$, and $v_p(a^2 + db^2)$ is even for every odd prime p with $(-d/p) = -1$.*

Proof. If $d > 0$, then clearly $a^2 + db^2 > 0$. If $d < 0$, then $-d > 1$ is squarefree, hence not a square, so $a^2 + db^2 \neq 0$. Let p be an odd prime with $(-d/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + db'^2$ by Lemma 2. Since $a^2 + db^2 = p^{2k}(a'^2 + db'^2)$, we get $v_p(a^2 + db^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{4|d|}$, or, if $d < 0$, with $n' \equiv -n \pmod{4|d|}$. Then $(-d/n') = (-d/n)$.*

Proof. Since d is squarefree, we can write $-d = \varepsilon 2^e q$ with $\varepsilon \in \{1, -1\}$, $e \in \{0, 1\}$ and q odd and positive. By multiplicativity and the reciprocity laws,

$$\left(\frac{-d}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since $4|d|$ is divisible by $4q$, and by $8q$ if $e = 1$, the claim for $n' \equiv n$ follows. If $d < 0$ and $n' \equiv -n \pmod{4|d|}$, then, as $\varepsilon = 1$, the factor (ε/n) does not change, $(2/n)$ does not change because $n'^2 \equiv n^2 \pmod{16}$, and each of $(-1)^{(n-1)(q-1)/4}$ and (n/q) is multiplied by $(-1)^{(q-1)/2}$, because $(n' - 1)/2 \equiv (n - 1)/2 + 1 \pmod{2}$ and $(n'/q) = (-1/q)(n/q) = (-1)^{(q-1)/2}(n/q)$. $\square$

Lemma 5. *Let n be a non-zero integer, and let s, u and m be positive integers such that every prime factor of s divides $2|d|$, both $2s$ and $4|d|s$ divide m , u is odd, $(-d/u) = -1$, and $n \equiv su \pmod{m}$. If $d > 0$, suppose also that $n > 0$. Then there is an odd prime p with $(-d/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by $4|d|$, n' is odd and $n' \equiv u \pmod{4|d|}$. If $d > 0$, then $n' > 0$, and if $d < 0$, then $|n'| \equiv \pm u \pmod{4|d|}$; in both cases Lemma 4 gives $(-d/|n'|) = (-d/u) = -1$. By multiplicativity, the product of $(-d/p)^{v_p(n')}$ over the prime factors p of $|n'|$ is -1 , so some prime factor p of n' satisfies $(-d/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 2d$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

2 The auxiliary polynomials

Put $F(x, y) = 2x^4 + xy + y^3 - y + 1$, so that (1) is the equation $F(x, y) = 0$. For the first identity, $d_1 = -11$, and we define

$$\begin{aligned} H_1 &= 2x - y + 18, \\ T_1 &= 8x^2 + 4xy + 146x + 2y^2 + 36y + 646, \\ U_1 &= 2x^2 + 4x + 149, \\ V_1 &= 4x - 31, \\ G_1 &= -2, \\ A_1 &= 8, \\ B_1 &= -4x - 39, \end{aligned}$$

for the second identity, $d_2 = 22$, and we define

$$\begin{aligned} H_2 &= 4x - 2y + 3, \\ T_2 &= 16x^2 + 8xy + 28x + 4y^2 + 6y + 5, \\ U_2 &= 4x^2 + 8x + 1, \\ V_2 &= 2x + 1, \\ G_2 &= -8, \\ A_2 &= -1, \\ B_2 &= 2x + 3, \end{aligned}$$

and, for the signs, we also use

$$Q_2 = 16x^2 + 8xy + 28x + 4y^2 + 6y + 5,$$

$$M_2 = 16x^4 + 64x^3 + 160x^2 + 104x + 23,$$

$$E_2 = 192x^2 + 352x + 44.$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 + 11V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 - 22V_2^2 = G_2F, \quad (3)$$

$$A_1U_1 + B_1V_1 = 7^4, \quad (4)$$

$$A_2U_2 + B_2V_2 = 2, \quad (5)$$

$$-8F = Q_2H_2 - M_2, \quad (6)$$

$$16Q_2 = (8x + 8y + 6)^2 + E_2. \quad (7)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) and (3) give

$$H_kT_k = U_k^2 + d_kV_k^2 \quad (k = 1, 2), \quad (8)$$

where the polynomials now denote their values at (x, y) .

Step 1: congruences. Checking the 968² residue pairs modulo 968, we find that (1) has exactly 396 solutions modulo 968, and that at each of them $H_1 \bmod 968$ lies in the set $\mathcal{R}_1$ listed in Table 1 or $H_2 \bmod 968$ lies in the set $\mathcal{R}_2$ listed in Table 1. For $k = 1$, each $r \in \mathcal{R}_1$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(11/u) = -1$; here for $s = 1$, $2s = 2$ and $44s = 44$ divide 968, and for $s = 11$, every prime factor of s divides 22, and $2s = 22$ and $44s = 484$ divide 968. For $k = 2$, each $r \in \mathcal{R}_2$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(-22/u) = -1$; here for $s = 1$, $2s = 2$ and $88s = 88$ divide 968, and for $s = 11$, every prime factor of s divides 44, and $2s = 22$ and $88s = 968$ divide 968.

Step 2: signs. *The value of H_1 .* By (4), U_1 and V_1 are not both zero, so $U_1^2 - 11V_1^2 \neq 0$ by Lemma 3 with $d = -11$, and (8) gives $H_1 \neq 0$.

The sign of H_2 . Since $F(x, y) = 0$, (6) gives $Q_2H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = 0$ we have $M_2 > 0$ and $E_2 > 0$. For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_2 > 0$ and $E_2 > 0$, so (7) gives $16Q_2 \geq E_2 > 0$, and $Q_2H_2 = M_2 > 0$ gives $H_2 > 0$.

Step 3: primes p with $(-d_k/p) = -1$. Let $k \in \{1, 2\}$, and let p be an odd prime with $(-d_k/p) = -1$, where $(d_1, d_2) = (-11, 22)$. We show that $v_p(H_k)$ is even. By (4)–(5), U_k and V_k are not both zero, so (8) and Lemma 3 show that $v_p(H_kT_k) = v_p(U_k^2 + d_kV_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 + d_kV_k^2$ by (8), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_kU_k + B_kV_k$. For $k = 1$, we have $R_1 = 7^4$. But p is odd, and $p \neq 7$ because $(11/7) = 1$, a contradiction. For $k = 2$, we have $R_2 = 2$. But p is odd, a contradiction. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_kT_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Step 1, there is k such that $H_k \equiv su \pmod{968}$ with s and u as in Lemma 5 for $d = d_k$. By Step 2, $H_k \neq 0$, and $H_k > 0$ if $d_k > 0$. Hence Lemma 5 applies to $n = H_k$ and $d = d_k$, so there is an odd prime p with $(-d_k/p) = -1$ such that $v_p(H_k)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The sets $\mathcal{R}_k$ of residues modulo 968 in Step 1.

$\mathcal{R}_1$, the residues of H_1					
3	173	337	497	663	823
13	179	339	513	673	833
29	189	349	515	689	839
31	205	355	525	691	849
41	207	365	531	701	865
47	217	381	541	707	867
57	223	383	557	717	877
73	233	393	559	733	883
75	249	399	569	735	893
85	251	409	575	745	909
91	261	425	585	751	911
101	267	427	601	761	921
117	277	437	603	777	927
119	293	443	613	779	937
129	295	453	619	781 = $11 \cdot 71$	953
135	297 = $11 \cdot 27$	469	629	789	955
145	305	471	645	795	965
161	311	481	647	805	
163	321	487	657	821	
$\mathcal{R}_2$, the residues of H_2					
5	161	337	513	689	865
17	181	357	533	709	885
37	193	369	545	721	897
69	213	389	565	741	917
73	245	421	597	773	949
77 = $11 \cdot 7$	249	425	601	777	953
93	269	445	621	797	
105	281	457	633	809	
125	301	477	653	829	
157	333	509	685	861	